

Adriano Machado

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About Me

I'm a recent M.Sc. in Artificial Intelligence graduate from the University of Porto. My thesis extended the NegotiationArena framework to open-weight language models, evaluating inference-time techniques such as Self-Refine and multi-agent deliberation for improving agent negotiation performance. As a student researcher, I studied how class imbalance affects synthetic tabular data generation, training and comparing deep generative models including DDPM, CTGAN, and TVAE.

Alongside my studies, I've spent the past few years volunteering across several student associations. I was part of the small team that redesigned UNI, the open-source mobile app used across the University of Porto, and more recently served as IT Manager at ESN Porto, supporting the 3500+ exchange students who pass through the city each semester.

Education

Faculty of Sciences, University of Porto

Porto, Portugal

M.Sc. IN ARTIFICIAL INTELLIGENCE

Sep. 2024 – Jul. 2026

- Final grade: 17/20
- Thesis: Evaluated the performance of open-weight LLMs in negotiation scenarios, evaluating the impact of Personas, Self-Refine and multi-agent deliberation. (Grade 18/20)
- Relevant Coursework: Natural Language Processing, Computer Vision, Deep & Reinforcement Learning.

Faculty of Engineering, University of Porto

Porto, Portugal

B.Sc. IN INFORMATICS AND COMPUTER ENGINEERING

Sep. 2021 – Jul. 2024

- Final grade: 15/20
- Relevant Coursework: Algorithms & Data Structures, Operating Systems, Software Design & Testing.

Escola Secundária da Lixa

Lixa, Portugal

HIGH SCHOOL, SCIENCE AND TECHNOLOGY

Sep. 2018 – Jun. 2021

- Final grade: 19/20

Work Experience

Faculty of Engineering of the University of Porto

Porto, Portugal

STUDENT RESEARCHER

Jul. 2025 – Oct. 2025

- Empirical study on how class imbalance in training data degrades the quality of fully synthetic tabular data.
- Trained and compared deep generative models (DDPM, CTGAN, TVAE) across varying levels of dataset imbalance.
- Worked with Python, Scikit-Learn, Synthcity, and pyMDMA.

Jumpseller

Porto, Portugal

INTERN

Feb. 2024 – Jun. 2024

- Worked on the automations feature of Jumpseller's platform (React/Redux)
- Built 12 dynamic, multilingual email templates using live store data (Ruby on Rails, Liquid).

Student Associations

Erasmus Student Network (ESN) Porto

Porto, Portugal

VOLUNTEER & IT MANAGER

Feb. 2025 – Aug. 2026

- Led the IT department of the Porto section of Europe's largest student association, which welcomes 3500+ exchange students each semester.
- Won the Lobos d'Ouro award for Best National IT Initiative.
- Built an internal asset requisition platform and a scholarship evaluation platform used to review hundreds of applications; built the section's new public website.
- Automated the generation of event descriptions for the public website and built a predictive model to forecast ESN Card demand.

SINF (Semana de Informática)

Porto, Portugal

HEAD OF THE PROGRAM DEPARTMENT

Jun. 2025 – Oct. 2025

- Led the department responsible for the program: 11 talks and 8 workshops, the largest program in six years, up from 4 talks and 2 workshops the previous edition.
- Coordinated speaker outreach, scheduling, and on-site logistics with a team of volunteers.

ENEI (National Meeting of Informatics Students)

Porto, Portugal

MEMBER OF THE PROGRAM DEPARTMENT

Sep. 2024 – Apr. 2025

- Recruited speakers and curated the program for a national student tech conference: 25 talks and 21 workshops.
- Handled outreach, scheduling, and speaker logistics.

- Member of the 3-person UI/UX team that led the redesign of UNI, an open-source mobile app used by University of Porto students.
- Member of the development team implementing the redesign and contributing to ongoing features.
- Worked with Flutter.

Projects

More projects and detailed write-ups are available at adrianomachado.pt.

Open-Weight LLMs as Negotiators (MSc Thesis)

[More Details](#)

LLMs · HUGGING FACE TRANSFORMERS, SLURM

2026

- Benchmarked 9 open-weight LLMs across Gemma, Mistral, and Qwen families (4B–27B) in 3 game-theoretic negotiation scenarios (Ultimatum, Trading, BuySell), extending NegotiationArena.
- Evaluated Self-Refine and team deliberation; Qwen had the best performance (63–67% win rate) and Self-Refine boosted disadvantaged seats by 5.6 percentage points.

Chessboard Photo to FEN

[More Details](#)

COMPUTER VISION · OPENCV, ULTRALYTICS YOLO, PYTORCH, OPTUNA

2025

- Built a digital twin pipeline that converts physical chessboard photographs into FEN notation.
- Hybrid pipeline (YOLOv8x + U-Net corner heatmaps) achieved 0.02 edit distance per board.
- YOLOv8x piece detection reached 0.995 mAP50.

Politeness Classification, from SVMs to LoRA

[More Details](#)

NLP · PYTORCH, HUGGING FACE TRANSFORMERS, W&B

2025

- Four-way politeness classifier on Intel's Polite Guard dataset, from classical ML baselines to fine-tuned transformers and parameter-efficient adapters.
- Fine-tuned RoBERTa with domain adaptation achieved 0.924 F1, matching the published state of the art and reducing large classification errors by 72% over SVM.
- LoRA adapters saved ~25% training time with minimal performance loss (0.901 F1), outperforming IA3 across all tested models.

Skills

Programming Languages	Python, Java, C, C++, JavaScript, Dart
AI & Data Science	PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Frameworks & Libraries	Flutter, React, Ruby on Rails
Tools & Platforms	Git, Docker, Linux, Firebase, Figma, SLURM, vLLM
Languages	Portuguese (native), English (fluent)